Work Sheet 3 (Data Structures)

1. Convert Following Infix expressions to postfix expressions. Show steps against each token/symbol.
   a. **Infix**:
      A + B * (C - D / E ^ F) * G

      Postfix:
      ABCDE/F^-*G*+

   b. **Infix**:
      ((A + B) * C - (D - E) ^ (F + G / H))

      Postfix:
      AB+C*DE-FGH/+^ -
   c. **nfix**:
      A + (B * C - (D / E ^ F) * G) * H

      Postfix:
      ABC*DEF^/G*-H*+
2. Convert Following Infix expressions to Prefix expressions. Show steps against each token/symbol.
   a. **Infix:**
      A * (B + C) / D
      Prefix:
      / * A + B C D
   b. **Infix:**
      A + B * (C ^ D - E) / F
      Prefix:
      + A / * B - ^ C D E F
   c. **Infix:**
      (A - B / C) * (D / E + F ^ G)

      Prefix:
      * - A / B C + / D E ^ F G

3. Evaluate/Execute following postfix/prefix expressions. Show steps against each token/symbol.
    a. 6 2 3 + - 3 8 2 / + * 2 ^ 3 +
    b. + * 6 - 4 2 / 8 4
    c. 2 3 ^ 5 4 * + 6 –
    d. * + 5 6 - 7 / 8 4
    e. 4 2 5 * + 1 3 2 * + /

4. **Implement the Stack ADT**
    a. **Define Stack Operations:**
        i. push(element)
        ii. pop()
        iii. peek()
        iv. isEmpty()
    b. **Choose Implementation Method:**
        i. Use an array as the underlying data structure.
        ii. Create dynamic array based on user's requirement.
    c. **Implement Redo Functionality**
    Redo Stack:
        i. When an action is undone, it is pushed onto the redo stack.
        ii. On redo command:
            1. Pop the last action from the redo stack.
            2. Reapply the action.
            3. Push it back onto the undo stack.

Example Workflow

1. **User Types "Hello":**
    - **Action:** Insert "Hello".
    - **Undo Stack:** [Insert "Hello"]
    - **Redo Stack:** []
2. **User Deletes "o":**
    - **Action:** Delete "o".
    - **Undo Stack:** [Insert "Hello", Delete "o"]
    - **Redo Stack:** []

3. **User Presses Undo:**
    - **Undo Stack Before:** [Insert "Hello", Delete "o"]
    - **Action:** Undo Delete "o".
    - **Undo Stack After:** [Insert "Hello"]
    - **Redo Stack:** [Delete "o"]
    - **Text:** "Hello"
4. **User Presses Redo:**
    - **Redo Stack Before:** [Delete "o"]
    - **Action:** Redo Delete "o".
    - **Redo Stack After:** []
    - **Undo Stack:** [Insert "Hello", Delete "o"]
    - **Text:** "Hell"

5. Open link and solve. A bonus marks of 2 can be judged in quiz for understanding problem. And 3 marks will be awarded on complete solution.

   https://onlinejudge.org/index.php?option=onlinejudge&page=show_problem&problem=455#google_vignette